ModelIt!

Teacher's Guide

Urban Heat Islands: Why Cities Are Hotter Than Forests

Modeling How Built Environments Amplify Extreme Heat

Quick Reference

Grade Level	Grade 09L2
NGSS Standard	HS-ESS3-4, HS-PS3-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS3-4, HS-PS3-4

Evaluate or refine a technological solution that reduces human impacts on natural systems. Plan and conduct an investigation to provide evidence that energy transfer through heating changes particle motion.

Three-Dimensional Learning

Science Practice	**Analyzing and Interpreting Data** — Students analyze temperature data across neighborhoods to identify patterns of heat inequality and evaluate intervention effectiveness.
Disciplinary Core Idea	**PS3.B Conservation of Energy and Energy Transfer** — Heat energy absorbed by dark surfaces is stored in thermal mass and radiated, creating measurable temperature differences between urban and rural environments.
Crosscutting Concept	**Cause and Effect** — Students model how the built environment creates feedback loops that amplify heat, and how targeted interventions can break these loops.

CAST Testing Alignment

MULTIPLE CHOICE:

Which intervention would provide the MOST cooling: (a) AC for all buildings, (b) 30 percent increase in tree canopy, (c) cool roofs on commercial buildings, (d) reducing traffic?

CONSTRUCTED RESPONSE:

Using your model, explain why a heat wave that is survivable in a suburban neighborhood with trees can be deadly in an urban neighborhood with concrete.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Urban heat islands are not just uncomfortable -- they are deadly. Heat kills more Americans than hurricanes, tornadoes, and floods combined. Understanding why some neighborhoods are dangerously hotter than others requires modeling the physics of the built environment.

The Physics of Urban Heat

Dark surfaces absorb up to 95 percent of incoming solar radiation and convert it to heat. Concrete and asphalt store this heat in their thermal mass and release it at night, preventing cooling. Vegetation cools through evapotranspiration, but cities have replaced trees with pavement.

The AC Feedback Loop

Air conditioning is both salvation and curse. It cools indoor spaces but pumps waste heat outdoors, adding 1-2 degrees to local air temperature in dense areas. Higher outdoor temperatures drive more AC use, which pumps more heat, creating a reinforcing loop.

Environmental Justice

Heat vulnerability maps closely mirror poverty and racial segregation maps. Historically redlined neighborhoods have 30-40 percent less tree canopy. Low-income residents are less likely to have AC, more likely to have outdoor jobs, and more likely to have health conditions exacerbated by heat.

Green Infrastructure Solutions

Trees provide shade AND evapotranspiration cooling. A single mature tree can provide the cooling equivalent of 10 room air conditioners. Cool roofs reflect solar radiation. Green roofs combine both. But these solutions take years to decades to mature.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify impervious surfaces, green space, ambient temperature, AC energy consumption, air quality, and health risk as the 6 components.

E **Establish Relationships**
Students map the reinforcing heat-AC loop and the environmental justice dimension of heat inequality.

V **Visualize & Model**
Students build a model comparing wealthy and underserved neighborhoods to reveal the structural roots of heat inequality.

E **Evaluate Outcomes**
Students run neighborhood comparison, heat wave, and green infrastructure scenarios.

R **Revise & Extend**
Students add cool roofs, transit, or nighttime temperature to explore additional cooling strategies.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"This map shows temperature. The red zones are 10+ degrees hotter. Who lives there?"*

Watch For

Show a real thermal map of a city. Ask students to guess which neighborhoods are hottest.

Slide 2: Learning Objectives

3 min

Say: *"Today we model why your neighborhood temperature depends on your zip code."*

Watch For

Pre-teach urban heat island and albedo.

Slide 3: Big Question

3 min

Say: *"And more importantly -- why are SOME neighborhoods hotter than others?"*

Watch For

Quick-write: What makes a place hot or cool?

Slide 4: LEVER Framework

2 min

Say: *"Six components with a sneaky reinforcing loop."*

Watch For

Preview the AC feedback loop concept.

Slide 5: Activity 1: Components

8 min

Say: *"Which can city planners change? Which are consequences?"*

Watch For

Sort components. Discuss why impervious surfaces and green space are policy choices.

Slide 6: Activity 2: Connect

10 min

Say: *"Find the reinforcing loop. Hint: AC is both a solution and a cause."*

Watch For

Map the heat-AC feedback loop. Discuss environmental justice dimension.

Slide 7: Activity 3: Simulate

10 min

Say: *"Compare two neighborhoods. One has trees, one has concrete. What happens during a heat wave?"*

Watch For

Students run scenarios and quantify the equity gap.

Slide 8: Discoveries

5 min

Say: *"Heat inequality is structural. It was designed into our cities."*

Watch For

Connect model findings to historical redlining and investment patterns.

Slide 9: STEM Challenge

12 min

Say: *"You have the budget. Where do you plant the trees?"*

Watch For

Groups design cooling plans prioritizing environmental justice.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Impervious Surface Area, Green Space Coverage
- **INTERNAL:** Ambient Temperature, Energy Consumption (AC), Air Quality Index, Heat-Related Health Risk

Relationships

- **Impervious Surface to Ambient Temperature:** POSITIVE (+) — More dark surfaces absorb more solar radiation, raising local temperature.
- **Green Space to Ambient Temperature:** NEGATIVE (-) — Vegetation cools through evapotranspiration and shade.
- **Ambient Temperature to Energy Consumption:** POSITIVE (+) — Higher temperatures drive more AC use.
- **Energy Consumption to Ambient Temperature:** POSITIVE (+) — AC pumps waste heat outdoors, creating a reinforcing loop.
- **Ambient Temperature to Air Quality:** NEGATIVE (-) — Heat accelerates ozone formation and traps pollutants.
- **Ambient Temperature to Health Risk:** POSITIVE (+) — Higher temperatures increase heat stroke, dehydration, and mortality risk.
- **Air Quality to Health Risk:** POSITIVE (+) — Poor air quality compounds heat-related health effects.

Key Discoveries

- Urban heat islands create a reinforcing loop: heat drives AC use which pumps more heat outside which drives more AC use
- The temperature difference between wealthy and poor neighborhoods can exceed 10 degrees F due to green space inequality
- Air conditioning is both a solution and a cause -- it cools indoor spaces while warming outdoor spaces and consuming energy
- Green infrastructure breaks the reinforcing loop by providing passive cooling through evapotranspiration

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Urban heat islands are just uncomfortable"	Urban heat kills more people than any other weather event. The 2003 European heat wave killed 70,000 people, mostly in cities. Heat strokes, dehydration, and cardiovascular events are medical emergencies.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"AC solves the heat problem"	AC cools individuals while warming the city. Every BTU removed from indoor spaces is pumped outdoors plus waste heat from the AC unit itself. Dense AC use can add 1-2 degrees to outdoor temperature, worsening the problem for those without AC.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Heat affects everyone equally"	Heat vulnerability is deeply unequal. Low-income residents are less likely to have AC, more likely to work outdoors, more likely to live in heat-absorbing neighborhoods, and more likely to have pre-existing conditions. Race and income predict heat death rates better than temperature does.